

UID EXPERIMENT-10

Name:Shanmuganathan S

Reg no.: 230701307

1. Set Up Your HTML File

- Create a new HTML file.
- Include the necessary HTML structure:
 - Use **<!DOCTYPE html>** to declare the document type.
 - Open the **<html>** tag with the language attribute set to "en".
 - Inside the **<head>** section:
 - Set the character set to UTF-8.
 - Add a viewport meta tag for responsive design.
 - Set the title of the document to "Inventory Management Visualization".
 - Add a **<style>** section to define basic styles for the body and canvas elements.
 - In the **<body>** section:
 - Add a heading **<h1>** for the title of the inventory management system.
 - Create two **<canvas>** elements for the pie chart and bar chart, specifying their IDs and dimensions.
 - Include the Chart.js library using a **<script>** tag with the source URL.
 - Link to an external JavaScript file (script.js) using another **<script>** tag.

HTML Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Inventory Management Visualization</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      text-align: center;

      margin: 50px;

    }

    canvas {

      margin: 20px auto;

    }

  </style>

</head>

<body>

  <h1>Inventory Management System</h1>

  <canvas id="pieChart" width="400" height="400"></canvas>

  <canvas id="barChart" width="400" height="400"></canvas>

  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

  <script src="script.js"></script>

</body>

</html>
```

2. Create the JavaScript File for Charts

- Create a new JavaScript file named **script.js**.
- Define the data for the inventory:
 - Create a constant object **inventoryData** with:
 - **labels**: An array of categories (e.g., 'Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys').
 - **datasets**: An array containing an object with:

- **label:** A string indicating what the data represents (e.g., 'Items in Stock').
- **data:** An array of numbers representing the quantity of items in stock for each category.
- **backgroundColor:** An array of color codes for each category.

3. Creating the Pie Chart

- Get the context of the pie chart canvas:
 - Use **document.getElementById('pieChart').getContext('2d')** to obtain the 2D rendering context.
- Create a new Chart instance for the pie chart:
 - Use the **Chart** constructor, passing in the context, chart type ('pie'), data (inventoryData), and options:
 - Set **responsive** to true for responsive resizing.
 - Define the title options to display the title "Inventory Distribution".

4. Creating the Bar Chart

- Get the context of the bar chart canvas:
 - Use **document.getElementById('barChart').getContext('2d')** to obtain the 2D rendering context.
- Create a new Chart instance for the bar chart:
 - Use the **Chart** constructor, passing in the context, chart type ('bar'), data (inventoryData), and options:
 - Set **responsive** to true for responsive resizing.
 - Define the title options to display the title "Items in Stock by Category".
 - Configure the scales to ensure the y-axis starts at zero by setting **beginAtZero** to true.

JAVA SCRIPT Code:

```
const inventoryData = {
  labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
  datasets: [{
    label: 'Items in Stock',
    data: [200, 150, 100, 80, 50],
    backgroundColor: [
      '#FF6384',
```

```
        '#36A2EB',  
        '#FFCE56',  
        '#4BC0C0',  
        '#9966FF'  
    ],  
    }  
};
```

// Creating the Pie Chart

```
const ctxPie = document.getElementById('pieChart').getContext('2d');  
const pieChart = new Chart(ctxPie, {  
    type: 'pie',  
    data: inventoryData,  
    options: {  
        responsive: true,  
        title: {  
            display: true,  
            text: 'Inventory Distribution'  
        }  
    }  
});
```

// Creating the Bar Chart

```
const ctxBar = document.getElementById('barChart').getContext('2d');  
const barChart = new Chart(ctxBar, {  
    type: 'bar',  
    data: inventoryData,  
    options: {  
        responsive: true,  
        title: {  
            display: true,
```

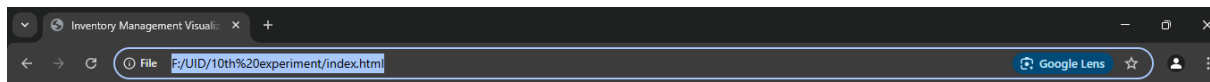
```
        text: 'Items in Stock by Category'
    },
    scales: {
        yAxes: [{
            ticks: {
                beginAtZero: true
            }
        }]
    }
}
});
```

5. Test the Implementation

- Open the HTML file in a web browser to view the charts.
- Ensure that both the pie chart and bar chart display correctly with the provided data.

6. Adjust and Customize (Optional)

- Modify the data, colors, or chart options as needed to better fit the inventory management system's requirements.



Inventory Management System

